
a Who is in each group
PAM on the % change in leg-press and leg-extension 1RM; gap selects K = 1, K = 2 used
Hi n = 8, Jaccard 0.88; Lo n = 7, Jaccard 0.84
PC2 (30%)
PC1 (70%)
HR LR
Hi Lo
b Proteins per contrast
1,900 tested; none at FDR < 0.05
Hi
Lo
Hi
Lo
T1
T2
T3
Training
Acute
proteins (down | up)
Down Π < 0.05
Up p < 0.05
Down p < 0.05
Up Π < 0.05
c Proteome
42 biopsies, 1,900 proteins
PERMANOVA
time R² 0.06, p = 0.0044
group R² 0.07, p = 0.43
PC2 (10.0%)
PC1 (12.9%)
Phenotype
30 biopsies, six size and strength traits
PERMANOVA
time R² 0.04, p = 0.037
PC2 (21.7%)
PC1 (61.0%)
T1 T2 T3
Hi Lo
d Pathways shared within each window
fgsea FDR < 0.05 in both groups of a window (interaction: either, top 5), shown in all nine contrasts; * opposite signs
Training
Acute
Hi - Lo
Interaction
Tricarboxylic acid cycle
Mitochondrial protein degradation
Mitochondrial gene expression
Regulation of muscle contraction
Protein localization
Muscle organ development
Eukaryotic translation elongation
Striated muscle contraction
RNA splicing via transesterification reac...
Dengue virus host interactions
Negative regulation of mRNA metabolic pro...
Fatty acid metabolism
Sarcomere organization
Organic acid catabolic process
Peroxisomal protein import
Regulation of striated muscle contraction
Generation of precursor metabolites and e...
Purine metabolism
Fatty acid metabolism
Cardiac chamber development
Protein RNA complex organization
ATP synthesis coupled electron transport
Mitochondrial protein degradation
Negative regulation of syncytium formatio...
Mitochondrial respiratory chain complex a...
Acyl CoA metabolic process
Proteasome: Negative regulation of hemopoiesis
Mitochondrial protein import
Mitochondrial translation
Regulation of insulin secretion
Striated muscle contraction
Modified amino acid metabolic process
Ribosomal small subunit biogenesis *
Proteasome: Negative regulation of hemopoiesis
Modified amino acid biosynthetic process
Hemostasis
Tight junction
Mitotic spindle
Hi Lo Hi Lo T1 T2 T3 Training Acute
NES
-log10 FDR
Database
Hallmark
Reactome
KEGG
GO BP
e Training signature rank
fry: each group's p < 0.05 training sets in the other group's ranking
up (n = 81), 38% same way, FDR 0.85
down (n = 53), 51% same way, FDR 0.85
up (n = 101), 62% same way, FDR 0.65
down (n = 74), 39% same way, FDR 0.65
Lo training sets in Hi's ranking
Hi training sets in Lo's ranking
rank in the other group's training contrast, up to down (n = 1900)
f Acute signature rank
fry: each group's p < 0.05 acute sets in the other group's ranking
up (n = 33), 79% same way, FDR 0.2
down (n = 49), 65% same way, FDR 0.23
up (n = 32), 84% same way, FDR 0.026
down (n = 34), 91% same way, FDR 0.092
Lo acute sets in Hi's ranking
Hi acute sets in Lo's ranking
rank in the other group's acute contrast, up to down (n = 190